

8.6

(a)

$$H_0: \sigma_M^2 = \sigma_F^2$$

$$H_1: \sigma_M^2 \neq \sigma_F^2$$

$$\alpha = 5\%$$

$$N_M = 577$$

$$df_M = 577 - 4 = 573$$

$$\hat{\sigma}_M^2 = \frac{SSEM}{df_M} = \frac{97161.9174}{573} = 169.5670$$

$$N_F = 1000 - 577 = 423, \quad df_F = 423 - 4 = 419$$

$$\hat{\sigma}_F^2 = 12.024^2 = 144.5766$$

$$\text{Test statistic: } F = \frac{\hat{\sigma}_M^2}{\hat{\sigma}_F^2} = \frac{169.5670}{144.5766} = 1.1728$$

$$RR: \{F: F > F_{0.025}(573, 419) = 1.186\}$$

$$\because 1.1728 < 1.186$$

$\therefore$ We do not reject H_0 . At the 5% significance level, there is no statistically significant difference in the variance of wages between males and females.

(b)

$$H_0: \sigma_{\text{SINGLE}}^2 = \sigma_{\text{MARRIED}}^2$$

$$H_1: \sigma_{\text{MARRIED}}^2 > \sigma_{\text{SINGLE}}^2$$

$$N_S = 400$$

$$df_S = 400 - 5 = 395$$

$$\hat{\sigma}_S^2 = \frac{SSE_S}{df_S} = \frac{56231.0382}{395} = 142.3571$$

$$N_M = 600$$

$$df_M = 600 - 5 = 595$$

$$\hat{\sigma}_M^2 = \frac{SSE_M}{df_M} = \frac{100703.0471}{595} = 169.2488$$

$$\text{Test statistic: } F = \frac{\hat{\sigma}_M^2}{\hat{\sigma}_S^2} = \frac{169.2488}{142.3571} = 1.1889$$

$$RR: \{ F: F > F_{0.05}(595, 395) = 1.16 \}$$

$$\because 1.1889 > 1.16$$

$\therefore$ We reject H_0 . At the 5% significance level, there is sufficient evidence to conclude that married individuals have more variable wages than single individuals.

(c)

$$\chi^2 = N \times R^2 \sim \chi^2_{(s-1)}$$

$$\text{Test statistic} = 59.03$$

$$\text{Critical value} = \chi^2_{0.05}(4) = 9.488$$

$$\therefore 59.03 > 9.488$$

$\therefore$ We reject H_0 . We can conclude that there is sufficient evidence of heteroskedasticity in the regression model at the 5% significance level.

The NR^2 test evaluates whether the variance of errors between observations in the regression model is constant, but it does not specifically examine whether the error variation differs between married and unmarried individuals. Therefore, despite the result of the NR^2 test suggests potential heteroskedasticity, it does not directly provide evidence regarding the specific issue discussed in part (b).

(d)

$$df = 4 + 4 + 6 - 2 = 12$$

$$\text{Critical value} = \chi^2_{0.05}(12) = 21.026$$

$$\text{Test statistic} = 78.82$$

$$\therefore 78.82 > 21.026$$

$\therefore$ We reject H_0 . We can conclude that there is statistically significant evidence of heteroskedasticity in the regression model at the 5% significance level.

(e)

$\text{robse} > \text{se} \Rightarrow \text{wider interval}$

$\text{robse} < \text{se} \Rightarrow \text{narrower interval}$

Intercept: $2.50 > 2.36 \Rightarrow \text{wider interval}$

EDUC: $0.16 > 0.14 \Rightarrow \text{wider interval}$

EXPER: $0.029 < 0.031 \Rightarrow \text{narrower interval}$

METRO: $0.84 < 1.05 \Rightarrow \text{narrower interval}$

FEMALE: $0.80 < 0.81 \Rightarrow \text{narrower interval}$

No, there is no inconsistency. It is a common misconception that White's heteroskedasticity-robust standard errors are always larger than the usual OLS standard errors. Depending on the exact nature of the heteroskedasticity in the sample, robust standard errors can be either larger or smaller than the usual OLS standard errors.

(f)

The addition of the intercept indicator variable MARRIED allows the expected wage, conditional on the explanatory variables EDUC, EXPER, METRO, and FEMALE to differ between those who are married and those who are not.

In part (b), we were asking about the amount of error variation between the two groups (married and single), which we found to be significant.

$\therefore$ The above two issues are quite different.

8.16

(a)

$$\text{MILES} = -391.55 + 14.20 \text{ INCOME} + 15.74 \text{ AGE} - 81.83 \text{ KIDS}$$

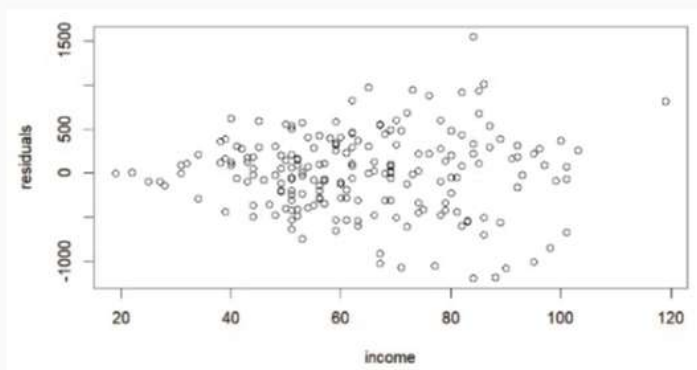
(se) (169.775) (1.800) (3.757) (27.130)

A 95% C.I. for B_4 is $(-81.83) \pm 1.972/4 \times 27.130$

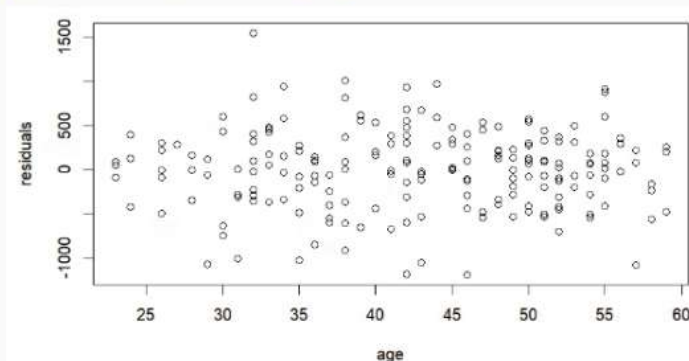
$$\Rightarrow [-135.32981, -28.32302]$$

$\therefore$ A 95% interval estimate for the effect of one more child on miles traveled is $[-135.32981, -28.32302]$

(b)



In the plot of the residuals against income, variation of the residuals increases as income increases.



In the plot of the residuals against age, the same effect is not apparent.

Overall, both plots are consistent with the presence of heteroskedasticity.

(c)

$$H_0: \sigma_1^2 = \sigma_2^2$$

$$H_1: \sigma_1^2 > \sigma_2^2$$

$$\alpha = 0.05$$

$$df = N - K = 90 - 4 = 86$$

$$\text{Test statistic: } F = \frac{\frac{\hat{\sigma}_2^2}{86}}{\frac{\hat{\sigma}_1^2}{86}} = \frac{\frac{315821.55}{86}}{\frac{101744.65}{86}} = 3.1041$$

$$RR: \{F: F > F_{0.95}(86, 86) = 1.4286\}$$

$$\because 3.1041 > 1.4286$$

$\therefore$ We reject H_0 . We can conclude that the error variance depends on Income.

(d)

Using heteroskedasticity robust standard errors, it has the same estimate for the effect of one more child on miles traveled, robust standard error is 29.15438 and the 95% interval estimate is $[-139.323, -24.32986]$.

Compare to part (a), this interval is wider because it has a larger standard error.

(e)

Set $\sigma_i^2 = \sigma^2 \text{INCOME}_i^2$, we need to multiply the model by $\frac{1}{\text{INCOME}}$. The GLS estimate for the effect of one more child on miles traveled is $-76,80629$, conventional GLS standard error is $21,84844$, and robust GLS standard error is $22,6186$.

⇒ The conventional GLS 95% C.I. is $[-119,8945, -33,71808]$
The robust GLS 95% C.I. is $[-121,4134, -32,1992]$

Compare to part (a) and part (d), these intervals are narrower because they have smaller standard errors.

Moreover, the conventional GLS interval is the narrowest interval because it has the smallest standard error.